

13 **C** Let Q be the heat supplied to the liquid per minute,

Considering the heat capacity of the liquid: $Q = mc\Delta T = mc(4)$

Considering the latent heat of the liquid: $Q(40) = mL$

Therefore,
$$\frac{c}{L} = \frac{\left(\frac{Q}{4m}\right)}{\left(\frac{40Q}{m}\right)} = \frac{1}{160} \text{ K}^{-1}$$

14 **D** The time required for the object to move from P to Q half a complete